Valuations and p-adic distance

Student handout

From factorials and prime counting to a new notion of closeness

7 problems | 100 points + 10 challenge points | Suggested time: 3–4 hours

Preparation. The lecture on Chebyshev's estimate and factorial/binomial valuations; course packet Chapters 7–8. No prior study of metric spaces is assumed.

Name: _____ **Due:** _____

Instructions. Write solutions on separate paper and justify your answers. You may use unique factorization, the binomial theorem, and integrality of binomial coefficients. You may use results of earlier problems even if you have not proved them. All logarithms are natural.

Notation. For a prime p and a nonzero integer m, $v_p(m)$ is the exponent of p in $|m|$. Set $v_p(0) = +\infty$, with $+\infty + t = +\infty$ for finite t. Recall that $\pi(x) = \#\{p \le x : p \text{ prime}\}$.

I. Factorials, binomial coefficients, and primes

1. Count the powers of a prime

[12 points]

(a) Prove Legendre's formula by counting multiples of $p, p^2, \ldots$:

$$v_p(n!) = \sum_{j \ge 1} \left\lfloor \frac{n}{p^j} \right\rfloor.$$

Explain why multiples of p^2 must be counted more than once and why the sum has only finitely many nonzero terms. (4)

(b) Compute $v_2(50!)$, $v_3(50!)$, and $v_5(50!)$. How many trailing zeros does $50!$ have in base ten? (4)

(c) Show that $v_p((pn)!) = n + v_p(n!)$. Deduce that, for $r \ge 1$,

$$v_p((p^r)!) = 1 + p + \cdots + p^{r-1} = \frac{p^r - 1}{p - 1}.$$

(4)

2. Why the middle of Pascal's triangle fits inside an lcm

[16 points]

Let $n \ge 1$ and $B_n = \binom{2n}{n}$.

(a) Use Problem 1 to show that

$$v_p(B_n) = \sum_{j \ge 1} \left(\left\lfloor \frac{2n}{p^j} \right\rfloor - 2 \left\lfloor \frac{n}{p^j} \right\rfloor \right).$$

Prove that every summand is either 0 or 1. *Hint:* Write $n/p^j = q + t$, with $q \in \mathbb{Z}$ and $0 \le t < 1$. (5)

(b) Compute $v_2\binom{20}{10}$, $v_3\binom{20}{10}$, and $v_5\binom{20}{10}$ without expanding the binomial coefficient. (3)

(c) Put $L_n = \operatorname{lcm}(1, 2, \ldots, 2n)$. Determine $v_p(L_n)$ and prove that $B_n \mid L_n$ by comparing valuations for every prime. (5)

(d) Prove that if $n < p \le 2n$, then $v_p(B_n) = 1$. (3)

3. Recover the scale of prime counting [16 points]

(a) Explain why the central coefficient is largest in row $2n$, and prove $\frac{4^n}{2n+1} \le B_n < 4^n$. *Hint:* Compare consecutive coefficients and sum the row. (4)

(b) Using Problem 2, prove

$$\log B_n \le \pi(2n)\log(2n), \qquad \pi(2n) \ge \frac{2n\log 2 - \log(2n+1)}{\log(2n)}.$$

Explain why monotonicity of π gives a constant $c > 0$ such that $\pi(x) \ge c\,x/\log x$ for all sufficiently large real x. (6)

(c) You may take the following lecture estimate as given: $\vartheta(x) = \sum_{p \le x} \log p \le Ax$ for $x \ge 2$, where $A > 0$ is a fixed constant. Split the primes at $\sqrt{x}$ to prove $\pi(x) \le \sqrt{x} + \frac{2Ax}{\log x}$ for $x \ge 4$. Deduce $\pi(x) \le Cx/\log x$ for some fixed C and sufficiently large x. (4)

(d) Do these bounds prove $\pi(x) \sim x/\log x$? Explain the distinction between $\asymp$ and $\sim$. (2)

II. Divisibility becomes a notion of distance

4. Extend the valuation to rational numbers [14 points]

For $a, b \in \mathbb{Z} \setminus \{0\}$, define $v_p(a/b) = v_p(a) - v_p(b)$; keep $v_p(0) = +\infty$.

(a) Prove that this definition is independent of the representation of a rational number as a fraction. (4)

(b) For nonzero $x, y \in \mathbb{Q}$, prove $v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x^{-1}) = -v_p(x)$. Compute $v_2(12/25)$, $v_5(12/25)$, and $v_3(9/14)$. (4)

(c) Prove, for all $x, y \in \mathbb{Q}$, that

$$v_p(x+y) \ge \min\{v_p(x), v_p(y)\}.$$

Show that equality holds when $v_p(x) \ne v_p(y)$. Give nonzero x, y for which the inequality is strict and $x + y \ne 0$. *Hint:* First work with integers, factor out the smaller power of p, then use a common denominator. (6)

5. The ultrametric inequality [14 points]

Define the *p-adic absolute value* by $|0|_p = 0$ and $|x|_p = p^{-v_p(x)}$ for $x \ne 0$. Define the *p-adic distance* by $d_p(x, y) = |x - y|_p$.

(a) Prove $|xy|_p = |x|_p|y|_p$ and the *ultrametric inequality*

$$|x+y|_p \le \max\{|x|_p, |y|_p\}.$$

When $|x|_p \ne |y|_p$, prove that equality holds. (6)

(b) Verify the defining properties of a metric: $d_p(x, y) \ge 0$, equality to 0 exactly when $x = y$, symmetry, and the triangle inequality. In fact, prove the stronger statement $d_p(x, z) \le \max\{d_p(x, y), d_p(y, z)\}$. (5)

(c) Compute $|81|_3$, $|1/27|_3$, and $|10|_3$. Which is closer to 0 in d_3: 81 or 1/27? Compare with ordinary absolute value. (3)

6. Closeness means agreement modulo a prime power [12 points]

(a) For integers a, b and $k \geq 0$, prove

$$d_p(a,b) \leq p^{-k} \quad \Longleftrightarrow \quad a \equiv b \pmod{p^k}.$$

(4)

(b) Compute $d_5(7,32)$, $d_5(7,132)$, and $d_5(7,8)$. Which of $32, 132, 8$ is closest to 7 in this metric? (3)

(c) List all integers b with $0 \leq b < 125$ and $d_5(7,b) \leq 5^{-2}$. Explain why the full set of integers satisfying this distance bound forms a congruence class. (3)

(d) Show that $|m|_p \leq 1$ for every integer m. Must this hold for every rational number? Give a reason or counterexample. (2)

III. First examples of p-adic convergence

Definition. A sequence (x_n) of rational numbers *converges p-adically* to $L \in \mathbb{Q}$ if, for every real $\varepsilon > 0$, there is an integer N such that $|x_n - L|_p < \varepsilon$ whenever $n \geq N$. Equivalently, the ordinary real sequence $|x_n - L|_p$ tends to 0. For integer x_n, L, Problem 6 says this happens exactly when, for every $k \geq 1$, eventually $x_n \equiv L \pmod{p^k}$.

7. Large integers can converge to zero [16 points]

(a) Prove $p^n \to 0$ p-adically. Use Problem 1 to prove $n! \to 0$ p-adically for every fixed prime p. In each case give a bound on $|x_n|_p$ that tends to 0. (5)

(b) Let $S_n = 1 + p + p^2 + \cdots + p^n$. Use the finite geometric-sum identity to prove

$$S_n \longrightarrow \frac{1}{1-p} \quad p\text{-adically.}$$

Compute the exact distance to the proposed limit. Specialize to $p = 2$: what is the limit of $1 + 2 + 4 + \cdots + 2^n$? (5)

(c) Let $T_n = \sum_{j=1}^{n} j!$. For any $m > n$, use the ultrametric inequality repeatedly to show

$$|T_m - T_n|_p \leq p^{-\lfloor (n+1)/p \rfloor}.$$

Deduce that (T_n) is *Cauchy*: for every $\varepsilon > 0$, there is N such that $|T_m - T_n|_p < \varepsilon$ for all $m, n \geq N$. You are not asked to find a rational limit. (4)

(d) Prove that a sequence of rational numbers has at most one rational p-adic limit. *Hint:* Apply the ultrametric inequality to $L - M = (L - x_n) + (x_n - M)$. (2)

Optional challenge: the geometry of an ultrametric [10 extra points]

(a) For three distinct rational numbers, prove that the largest of their three pairwise p-adic distances occurs at least twice. Thus every triangle is isosceles, with its two largest sides equal (equilateral triangles are allowed). (4)

(b) For a real $r > 0$, define the closed ball $B(a,r) = \{x \in \mathbb{Q} : d_p(x,a) \leq r\}$. Prove that if $b \in B(a,r)$, then $B(b,r) = B(a,r)$. (3)

(c) Prove that two closed balls are either disjoint or one is contained in the other. *Hint:* If the balls intersect, choose a point in their intersection and apply part (b). (3)